DROPDOWN ACADEMY

Aester EQ

User manual

Version 1.3.0 — DropDownAcademy edition

24-band equalizer with static, dynamic and spectral processing

macOS Apple Silicon · Windows x64

www.dropdownacademy.com

Contents

1. What Aester EQ is
2. Requirements and installation
3. First launch
4. The interface
5. Bands
6. Static, Dynamic, Spectral
7. Stereo placement
8. Engine, phase and latency
9. Mix
10. Solo
11. Analyzer and Freeze
12. Output, balance and Mid/Side
13. Presets, Undo/Redo, A/B
14. Practical examples
15. What the plugin does not do
16. Troubleshooting
17. Specifications and license

1. What Aester EQ is

Aester EQ is a 24-band equalizer built for targeted work: few bands, precise control, and a per-band choice between fixed gain (Static), level-driven gain (Dynamic) and frequency-selective processing (Spectral).

The static path uses linear filters: gain and bandwidth determine the tonal change, with no intentional saturation stage. Dynamic mode reads a level detector and moves filter gain; Spectral mode works on a short-time transform to act on frequency regions.

The plugin ships as Audio Unit and standalone on macOS, and as VST3 and standalone on Windows.

2. Requirements and installation

Requirements

macOS	macOS 11 or later, Apple Silicon (arm64). Formats: AU, standalone.
Windows	Windows 10/11 64-bit. Formats: VST3, standalone.
Host	Any DAW that loads AU (macOS) or VST3 (Windows).

macOS

1. Save your project and close the DAW.
2. Open `Aester-EQ-DropDownAcademy-1.3.0-macOS-arm64.dmg`.
3. Run `Aester-EQ-DropDownAcademy-Installer.pkg` and complete the installation (administrator password required).

4. The package installs `Aester EQ.component` in `/Library/Audio/Plug-Ins/Components` and `Aester EQ.app` in `/Applications`.
5. Restart the DAW and look for **DropDownAcademy → Aester EQ**.

> The package is ad hoc signed and not notarized. If macOS blocks the first launch of the standalone, right-click the app, choose **Open** and confirm. No driver, system extension or background service is installed.

Windows

1. Save your project and close the DAW.
2. Run `Aester-EQ-DropDownAcademy-1.3.0-Windows-x64.exe` and allow the administrator prompt.
3. The VST3 is installed in `C:\Program Files\Common Files\VST3\Aester EQ.vst3`; the standalone in `C:\Program Files\DropDownAcademy\Aester EQ`.
4. Restart the DAW and rescan plug-ins.

> The installer is unsigned, so SmartScreen may ask for confirmation. If the host reports missing DLLs, install the official *Microsoft Visual C++ 2015-2022 Redistributable (x64)* from Microsoft, never individual DLL files from third-party sites.

Updating from 1.2.0

Version 1.3.0 uses a new product identity (bundle identifier and manufacturer code). The previous version is not overwritten: remove the old `Aester EQ.component` (or `Aester EQ.vst3`) and the old application, then install again and rescan. Parameters and states saved with 1.2.0 load correctly, but some hosts may report the previous instance as missing because the plug-in identity changed.

Uninstalling

- **macOS:** delete `Aester EQ.component` from `/Library/Audio/Plug-Ins/Components` and `Aester EQ.app` from `/Applications`.
- **Windows:** use *Apps and features*; the uninstaller removes the VST3, the standalone and the shortcuts.

User presets and session files are not deleted.

3. First launch

A new instance opens **without active bands**. The graph shows *Double-click the graph to add a band*: double-click anywhere to create band 1, or press **+ Add band**.

The window starts at 1000 × 700 points. Resize it by dragging the lower-right corner or use the **Size** menu. The chosen size is remembered by the instance while it stays loaded.



Main window: response graph, floating panel for the selected band, footer with global controls.

4. The interface

Area	What it contains
Top bar	Product name, Undo/Redo, preset menu, A/B, Copy, Help, Size.
Graph	*Wet* frequency response before Mix, band nodes, analyzer. Labelled *WET RESPONSE / FREQUENCY*.
L / R meters	Right of the graph, per-channel peak.
Floating panel	Appears when a band is selected: FREQ, GAIN, Q, SOLO, shape, slope, stereo placement, mode. It stays still while a knob is turned and follows frequency when released.
Band strip	Numbered cards (01–24) over four pages of six. Each card has an enable switch and numeric fields for frequency, gain and Q.
Footer	Processing path, analyzer, Freeze, dB range, L/R–Mid/Side selector, INV ALL and the BALANCE, MIX, OUTPUT rotaries.

5. Bands

Create, select, edit

- **Double-click** the graph: creates a band at that point.
- **Single click** a node: selects the band and opens the floating panel.
- **Drag** a node: sets frequency and gain.
- **Scroll** on the graph: changes the selected band's Q.
- **Double-click** a knob in the panel: resets that parameter to its default.
- **Right-click** a node: band actions.
- **Del**: disables the selected slot.

Numeric fields on the cards allow exact values. The floating panel offers the same three controls as knobs, with frequency and gain also shown next to the node.

Available shapes

Shape	Typical use
Bell	Symmetric bell: the most common tonal correction.
Low Shelf	Constant gain below the chosen frequency.
Low Cut	High-pass with selectable slope.
High Shelf	Constant gain above the chosen frequency.
High Cut	Low-pass with selectable slope.
Notch	Narrow cut at a precise frequency.
Band Pass	Passes only the region around the frequency.
Tilt Shelf	Progressive balance between lows and highs.
Flat Tilt	Linear tilt of the response.
All Pass	Changes phase without changing magnitude.

Cut slopes

The slope menu is active for Low Cut and High Cut only: 6, 12, 24, 36, 48, 72 or 96 dB/octave. On the other shapes the control is disabled.

24 bands over four pages

Cards are grouped in four pages of six. The arrows next to the page indicator switch pages; the counter at the bottom left reports how many bands are active out of the total (for example *3 / 24 BANDS*).

6. Static, Dynamic, Spectral

Static

Band gain is fixed: classic equalization. In Live mode, Static adds no latency.

Dynamic

Filter gain is driven by signal level. The expanded panel shows:

THRESHOLD	Trigger level. The current detector value is displayed next to it.
RANGE	How far gain moves: negative reduces level when the threshold is exceeded, positive expands it.
ATTACK	How quickly gain reaches the triggered value.
RELEASE	How quickly it returns.
Trigger Band / Free	**Band**: the detector follows the band's frequency region. **Free**: allows separate low and high detection cuts.
External key	Each band has its own switch: it only works if the DAW also routes a sidechain source. A silent or missing external key is not treated as the main input.

Spectral

Analysis runs per frequency: gain is applied to short-time spectral regions. It adds latency compared with the Live path (see chapter 8) and is meant for selective work that IIR filters cannot do.

> As of 1.3.0 these are not implemented: detector audition, automatic threshold and a separate Spectral Free detector.

7. Stereo placement

Stereo	Applies the band to both channels.
Left / Right	Acts on one channel only.
Mid / Side	Acts on the sum (Mid) or difference (Side) component.

On a mono signal, processing that only addresses Right or Side has no effect. A band's stereo placement is respected by Solo as well.

8. Engine, phase and latency

Mode	Behaviour	Added latency
Live	IIR filters. No added delay in Static and Dynamic.	0 samples
Analog FIR	Samples the complex filter response at an elevated design rate.	320 samples
Linear FIR	Keeps magnitude with a symmetric kernel; five resolutions (Low, Medium, High, Very High, Maximum).	Increases with resolution
Spectral	Short-time analysis and synthesis.	1024 samples on top of the chosen path

Higher FIR resolutions improve low-frequency definition but increase latency and pre-ringing. The latency value in the footer is what the plugin reports to the host and uses to align the dry signal in Mix.

9. Mix

Mix is a linear, latency-aligned blend between dry and processed audio: 100% is wet only, 0% is dry only. Changes are smoothed over 20 ms.

- The graph shows the **wet response before Mix**: it does not plot the phase-dependent cancellations that a dry/wet sum can produce.
- A 50% Mix is **not** the same as halving every band's gain in dB, because complex filter responses sum.

10. Solo

SOLO temporarily isolates the selected band's frequency region from the already processed output (it does not bypass the other bands). Audition width follows the band's Q: bell, notch, band-pass, tilt and all-pass shapes use a unity-centre band-pass monitor; Low Shelf and High Cut audition the low region, High Shelf and Low Cut the high region.

- Respects the band's stereo placement.
- Follows processing latency and uses a 20 ms crossfade.
- When the selected band changes, the monitor fades out before switching.
- It is temporary: not automatable, not stored in presets, and it ends when the editor closes or the band is disabled.

While Solo is active, only the auditioned band stays highlighted in yellow; the other bands and the combined response are dimmed.

11. Analyzer and Freeze

Analyzer selects Off, Pre (before processing), Post (after) or Pre+Post. Its context menu adds:

- **Display tilt**: 0, 3, 4.5 or 6 dB/octave, for easier spectrum reading.
- **Side Chain spectrum**: also shows the external key spectrum.

Freeze holds the spectrum display: audio, gain reduction and meters keep running. **Resume** restarts display updates. Freeze is editor-local and is not a stored plugin parameter.

The adjacent button (12 dB / 30 dB / 6 dB) changes the vertical range. Output meters use peak retention with time-based decay.

12. Output, balance and Mid/Side

Footer rotaries

OUTPUT	Output gain on the wet path. Double-click resets to 0 dB.
MIX	Dry/wet blend (chapter 9). Double-click resets to 100%.
BALANCE	Post-Mix balance. Shows L/R percentages or M/S with *C* at centre. Double-click recentres.

Because balance and polarity sit **after** Mix, they stay active even at 0% Mix. Plugin bypass restores the unbalanced, non-inverted signal.

Mid / Side output

Selecting **Mid / Side** from the stereo mode menu opens the *MID / SIDE OUTPUT* panel with:

- independent **MID** and **SIDE** gain;
- **INV MID** and **INV SIDE** polarity switches;
- **INV ALL**, which inverts the final stereo signal.

The centre balance knob favours Mid to the left and Side to the right. These controls act after Mix, require a stereo signal, are stored in presets and are released by bypass. In L/R mode the panel is hidden.

> The output control is not a limiter: it does not protect against peaks introduced by equalization.

13. Presets, Undo/Redo, A/B

Preset menu	Save and load state, reset parameters, toggle bypass.
Undo / Redo	Arrows in the top bar. They cover changes made in the open editor.
A* / B	Two comparison states local to the editor. The asterisk marks the active side.
Copy	Copies the current state to the plugin clipboard.
Size	Preset window dimensions.

Undo/Redo history and A/B are editor-local: they are separate from preset and session state persistence.

14. Practical examples

Vocals: clean without hollowing

1. Low Cut at 70–90 Hz, 24 dB/octave.
2. A narrow bell (Q 3–6) cutting the most annoying resonance.
3. A gentle High Shelf for air, if needed.
4. A Dynamic band cutting the region that stings when the level rises, moderate negative range.

Drum bus: transient control

1. Low-Q bell boost on the body (60–120 Hz).
2. A Dynamic band cutting mid-highs to tame hard hits.
3. Watch the L/R meters and use Output to return to the starting level.

Master: surgical corrections

1. Use few bands and small gains.
2. If you use Linear FIR for linear phase, account for latency when comparing with dry, and use Mix with care.
3. Always check against bypass: the correction must remain an improvement.

15. What the plugin does not do

- It is not a limiter and does not normalize level.
- It does not claim perceptual transparency: it is an equalizer, with the phase and pre-ringing of the chosen modes.
- These packages do not include AAX.

- Solo is not automatable and is not stored in presets.
- The external key is not managed by the plugin alone: the DAW must route it.

16. Troubleshooting

Symptom	What to do
The plugin is not listed in the DAW	Restart the DAW after installation and force a plug-in rescan. On macOS check that `Aester EQ.component` is in `/Library/Audio/Plug-Ins/Components`.
macOS blocks the standalone	Right-click the app → **Open**. The package is not notarized.
Windows flags the file as unsafe	Confirm in SmartScreen: the installer is not signed with a commercial certificate.
Missing DLL error on Windows	Install the official Microsoft Visual C++ 2015-2022 Redistributable x64.
The session cannot find the plugin after updating	1.3.0 has a new identity: reinstall 1.2.0 to reopen the session, or reload the plugin and restore the parameters.
Latency feels high	Check the phase path in the footer: Live is zero delay, Analog FIR 320 samples, Spectral +1024.
A/B comparison does nothing	A/B is local to the open editor and is not stored in presets.

17. Specifications and license

Product	Aester EQ 1.3.0 — DropDownAcademy
Bands	24, over four pages of six
Shapes	10 (Bell, Low Shelf, Low Cut, High Shelf, High Cut, Notch, Band Pass, Tilt Shelf, Flat Tilt, All Pass)
Modes	Static, Dynamic, Spectral
Stereo placements	Stereo, Left, Right, Mid, Side
Paths	Live IIR, Analog FIR, Linear FIR (5 resolutions), Spectral
Formats	AU + standalone (macOS arm64), VST3 + standalone (Windows x64)
License	AGPL-3.0-only. Source code is included in the macOS package.

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